HACKCULT ’26 · IIT GANDHINAGAR · TRACK 2 · HC26-CULT01

HeritageLens

India’s field archives, read, mapped and rebuilt.

An AI platform that reads handwritten Gujarati, Hindi and Sanskrit field notes, translates them, pins every place they mention on a map and turns site photographs into explorable 3D digital twins.

Team Binary Duo

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Rudra Umra
SNPIT&RC, Umrakh, Surat

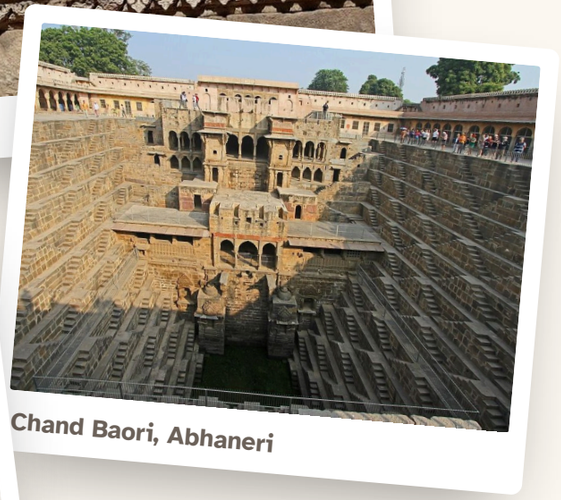
Live app: heritagelens-hc26.vercel.app

Code: github.com/kushpatel2601/HeritageLens

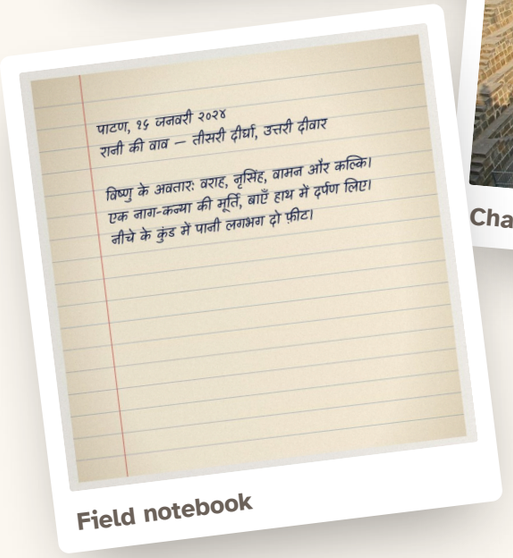
Demo video (3 min)



Rani ki Vav, Patan



Chand Baori, Abhaneri

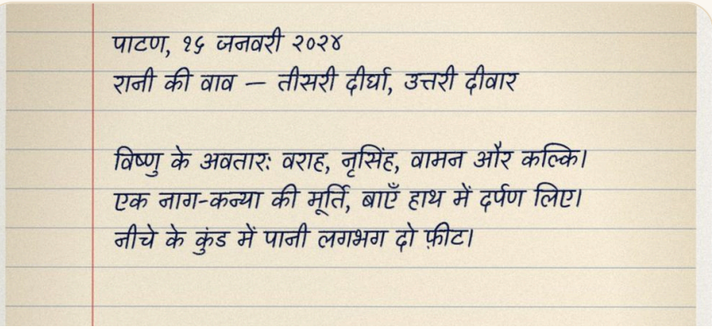


Field notebook

Photographs: Wikimedia Commons contributors (CC BY / CC BY-SA)

PROBLEM SOLVED

India’s heritage lives where search can’t reach



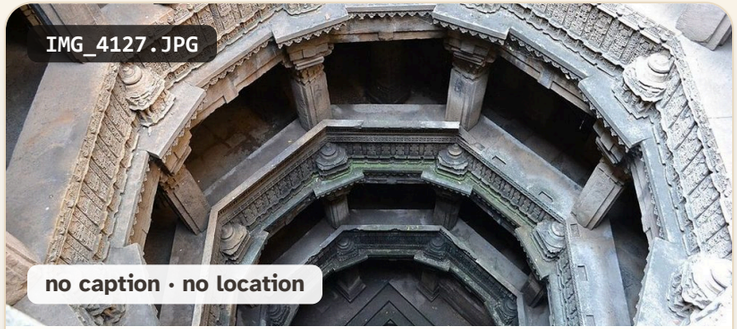
Field journals

- Handwritten in ગુજરાતી, हिन्दी and Sanskrit
- One place, many spellings: વાવ · बावड़ी · vav



Scanned PDFs and decks

- Text is locked inside page images
- No OCR, no index, no translation



Untagged photographs

- Thousands of images with no caption or place
- Nothing links a photo or a note to its site

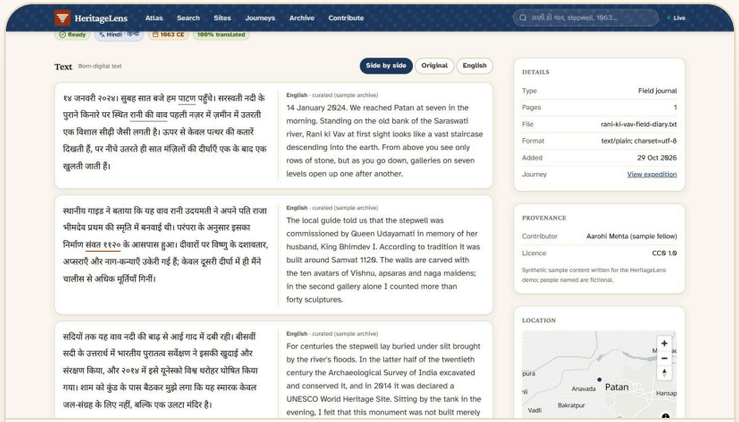
Decades of fieldwork that nobody can search, read across scripts or find on a map.

HC26-CULT01 asks for three pathways

1. **Read regional text archives** →
2. **Map them on an interactive atlas** →
3. **Rebuild sites as 3D digital twins**

SOLUTION

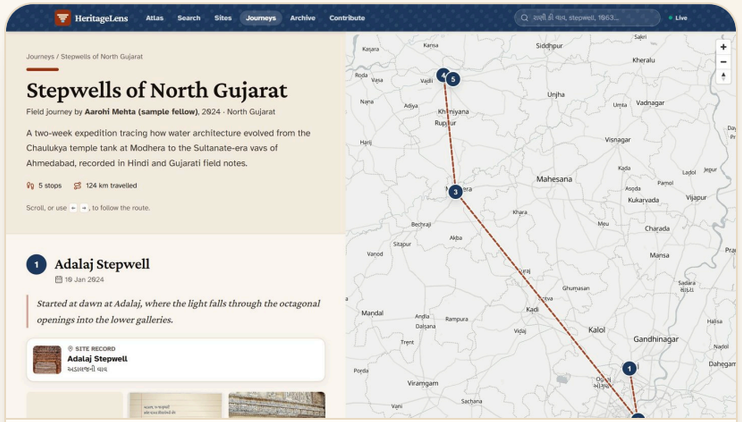
One platform covers all three pathways, live today



PATHWAY 1

Regional text archives

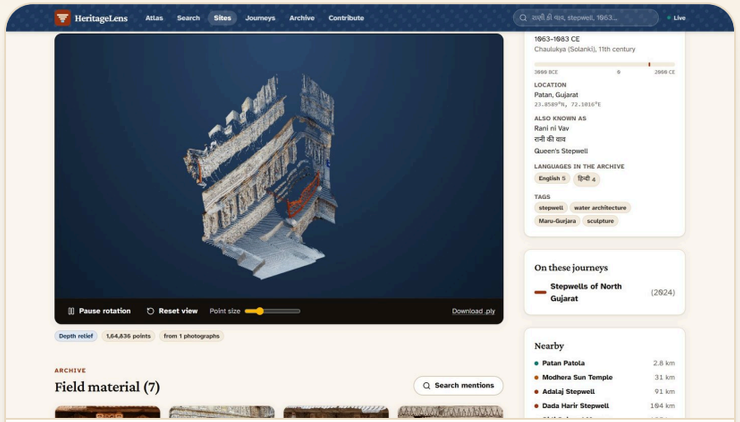
- OCR for Gujarati, Hindi, Marathi, Sanskrit and English
- Translated to English and read side by side
- Places, sites and dates pulled out and linked



PATHWAY 2

Interactive atlas

- 17 sites, 3 fellowship journeys, 59 geotagged items
- Geotags from EXIF, the linked site or places named in the text
- Journeys play as a step-by-step story



PATHWAY 3

3D digital twins

- 1 photo → Depth Anything V2 relief (165k points)
- 8+ photos → Gaussian splat (COLMAP + gsplat)
- Built on a free Colab GPU, viewed in the browser

ALSO LIVE

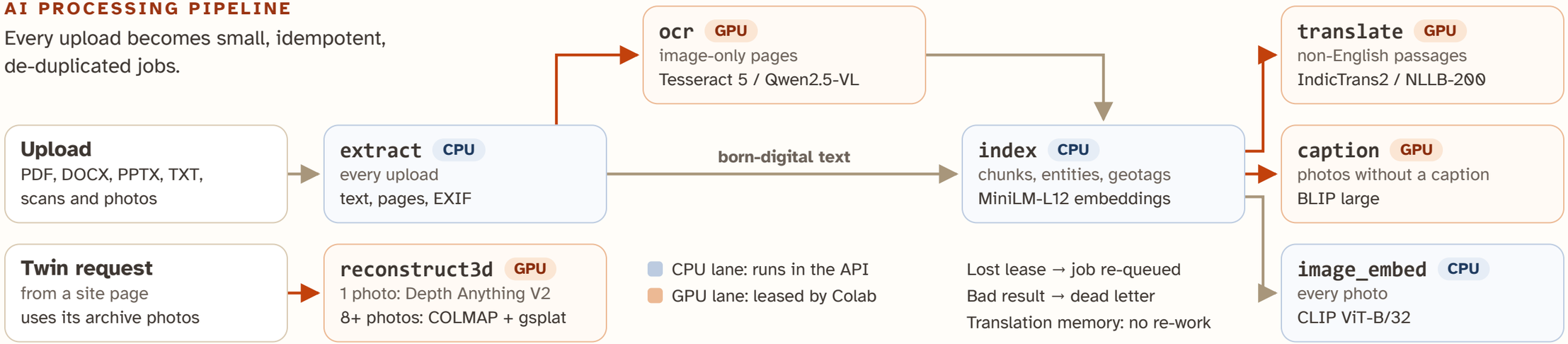
Cross-script search · **Live pipeline dashboard** · **Drag-and-drop uploads** · **Open REST API** · **Works on phones**

HOW IT WORKS

AI pipeline in, cross-script search out

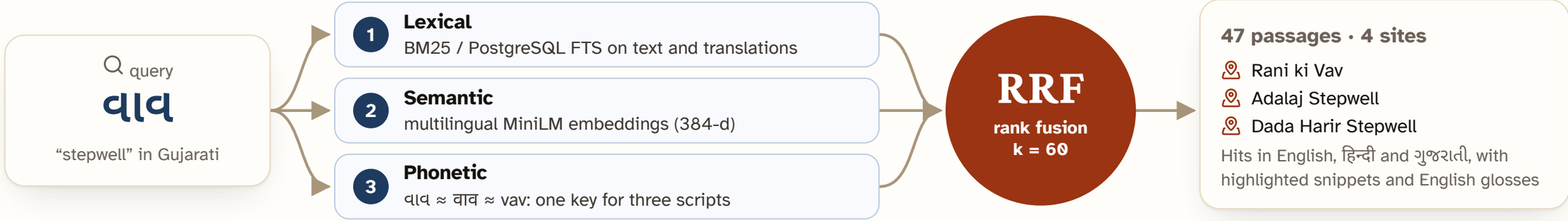
AI PROCESSING PIPELINE

Every upload becomes small, idempotent, de-duplicated jobs.



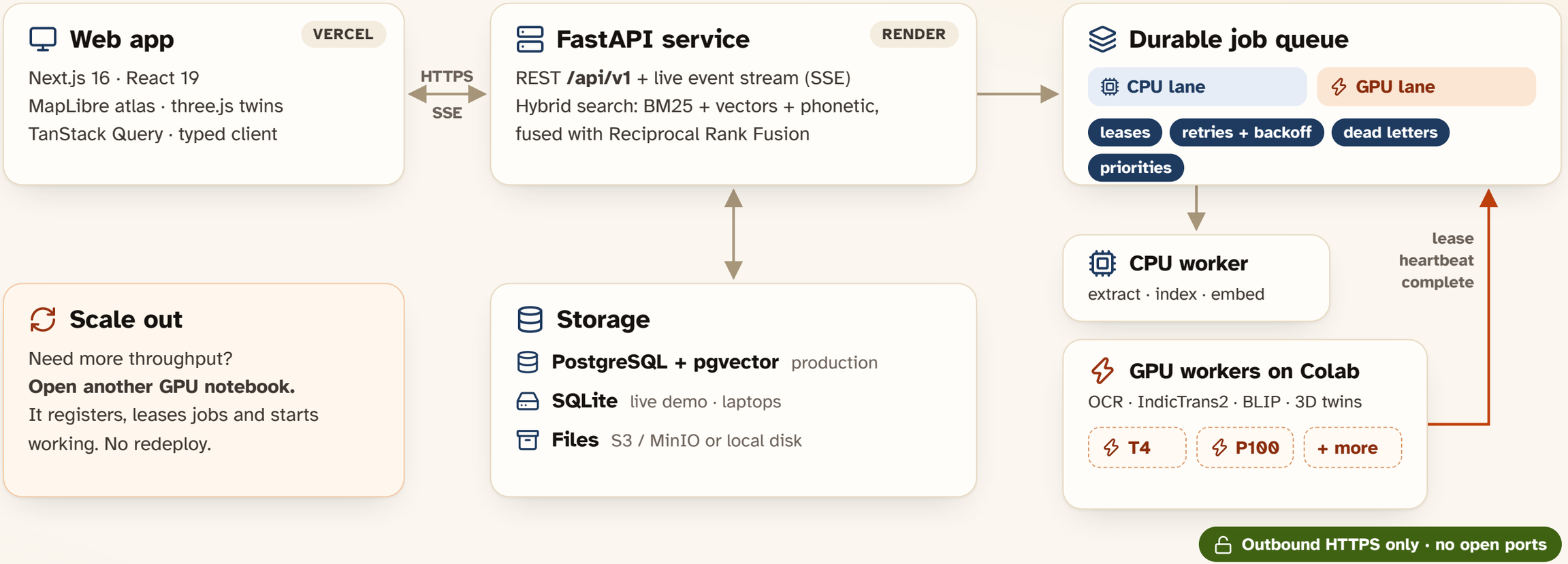
HYBRID CROSS-SCRIPT SEARCH

Three retrievers, fused with Reciprocal Rank Fusion (k = 60).



SYSTEM ARCHITECTURE

Stateless API, durable queue, GPUs that pull work



Stateless API
Add replicas behind a load balancer; no session state.

SKIP LOCKED leasing
Many workers share one queue without doing work twice.

Leases + heartbeats
A Colab disconnect re-queues the job automatically.

Live updates over SSE
Pipeline progress streams straight to the browser.

DATABASE SCHEMA

PK primary key → table foreign key vec pgvector + HNSW FTS full-text

Ten tables, one model for SQLite and PostgreSQL

ATLAS

journeys

id	PK
slug	unique
title, fellow	
year, region	
summary	
color	

sites

id	PK
slug	unique
name, name_native	
category, state	
lat, lon	
era_start, era_end	
phonetic	
embedding	vec 384
cover_asset_id	→ assets

journey_stops

id	PK
journey_id	→ journeys
site_id	→ sites
seq	unique/journey
lat, lon	
visited_on	
narrative	

twins

id	PK
site_id	→ sites
method	relief · splat
format	ply · splat
point_count	
source_asset_ids	
storage_key	

ARCHIVE

assets

id	PK
kind	5 kinds
sha256	unique
site_id	→ sites
journey_id	→ journeys
parent_id	→ assets
lat, lon, geo_source	
license, caption	
image_embedding	vec 512

chunks

id	PK
asset_id	→ assets
text	FTS
language, script	
translation	
translation_engine	
phonetic	
embedding	vec 384

pages

id	PK
asset_id	→ assets
page_no	
text	
source	native · ocr
ocr_engine	
ocr_confidence	

entities

id	PK
asset_id	→ assets
chunk_id	→ chunks
site_id	→ sites
kind	site · place · date
normalized	
lat, lon, year	

PIPELINE

jobs

id	PK
type, queue	cpu · gpu
status, priority	
dedupe_key	
asset_id	→ assets
site_id	→ sites
attempts, run_after	
lease_owner	
lease_expires_at	

workers

id	PK
name, queue	
device	
capabilities	
last_seen_at	
jobs_done	
jobs_failed	

SQLAlchemy 2 models with Alembic migrations. On PostgreSQL: HNSW cosine indexes on all three embedding columns, a GIN full-text index on chunks, and a `(queue, status, priority, id)` index that GPU and CPU workers lease from with `FOR UPDATE SKIP LOCKED`.

TECH STACK

Open-source end to end, typed from database to UI

Frontend
Map-first, accessible web app

- Next.js 16
- React 19
- TypeScript
- Tailwind CSS v4
- TanStack Query 5
- MapLibre GL + OpenFreeMap
- three.js · React Three Fiber
- openapi-fetch (generated)

Backend
Async API, queue and workers

- Python 3.12
- FastAPI
- Pydantic v2
- SQLAlchemy 2 (async)
- Alembic
- Server-Sent Events
- OpenAPI 3.1

Data and search
Hybrid retrieval over every script

- PostgreSQL + pgvector
- SQLite
- BM25 · PostgreSQL FTS
- HNSW vector indexes
- Reciprocal Rank Fusion
- Phonetic cross-script keys
- S3 / MinIO

AI models
Read, translate, describe, embed

- IndicTrans2
- NLLB-200 (fallback)
- Tesseract 5
- Qwen2.5-VL-3B (OCR)
- BLIP large captions
- multilingual MiniLM-L12 (ONNX)
- CLIP ViT-B/32

3D and GPU
Twins from photos the archive holds

- Depth Anything V2
- COLMAP
- gsplat (Gaussian splats)
- PLY point clouds
- Colab T4 / P100
- heritage-gpu worker package

Delivery and quality
Free-tier hosting, CI on every push

- 326 automated tests
- pytest · Vitest
- ruff · mypy · ESLint · tsc
- GitHub Actions
- Vercel
- Render (Docker)
- UptimeRobot

VALUE AND IMPACT

Dormant documentation becomes a living, searchable archive

Researchers and students

Find every journal page, photo and scan about a site in one search, in English, Gujarati or Hindi, with the source page one click away.

Archivists and institutions

Drop in PDFs, decks and photos. OCR, translation, captions, geotags and 3D twins run automatically, with provenance and licence kept on every item.

Heritage tourism and the public

Follow a fellow's journey across the map, read their notes in your own script, and walk around a site in 3D before you visit.

● LIVE TODAY ON THE DEPLOYED SITE

17	3	59	153	13	3	306	326
heritage sites	fellowship journeys	archive items	linked entities	3D digital twins	languages and scripts	pipeline jobs completed	automated tests

₹0

to run today: Vercel and Render free tiers, with free Colab T4 / P100 GPUs doing the heavy AI work. A small archive can adopt it without a budget line.

Fit to HC26-CULT01

- **All three pathways live:** contextual reader, journey map, 3D twins.
- **Every answer cites its source** page, photo or scan, with licence and attribution.
- **Search across scripts:** type વાવ, find "stepwell" and बावड़ी too.
- **Open REST API** with OpenAPI docs, so other portals can build on it.

POTENTIAL SCALABILITY

Every layer scales out without a rewrite

LAYER	LIVE DEMO TODAY	HOW IT SCALES
Web	Next.js on Vercel's CDN	Static and server-rendered pages are served from the edge; no change needed.
API	One FastAPI container, 512 MB	Stateless: add containers behind a load balancer. Config is all environment variables.
Job queue	Jobs table in the database	Workers lease with `FOR UPDATE SKIP LOCKED`, so many can pull at once without double work.
CPU work	Worker inside the API process	Turn it off and run `python -m app.cli worker` on as many machines as the backlog needs.
GPU work	One Colab T4 / P100 notebook	Any GPU anywhere joins with the API URL and a token: Colab, a lab PC or a cloud spot instance.
Database	SQLite file	Same models on PostgreSQL + pgvector with HNSW indexes; add read replicas for search traffic.
Files	Container disk	Storage is behind one interface; switch to S3 or MinIO and put thumbnails behind a CDN.

BUILT IN

- **Idempotent jobs (dedupe keys)**
- **SHA-256 de-duplication**
- **Translation memory**
- **Leases + heartbeats**
- **Retries, backoff, dead letter**

Today · demo
1 API, 1 GPU notebook, SQLite. Runs for ₹0 on free tiers.

→

One institution
Managed PostgreSQL, S3, a few CPU workers and a GPU worker for a museum or university archive.

→

National network
Partner archives feed one shared index; GPU workers grow and shrink with the queue.

LIMITATIONS AND ROADMAP

What it can't do yet, and what comes next

⚠ Honest limitations

Free hosting. The API sleeps after 15 idle minutes (a keep-alive monitor covers this) and its disk resets on restart, so new uploads are temporary.

GPU jobs wait for a GPU. Without a Colab worker, handwriting OCR, captions and splats queue up; the Tesseract fallback is weak on handwriting.

Search by image is off on the 512 MB server (CLIP needs more memory).

Small, curated dataset. 17 sites and 59 items; the sample field journals are synthetic and their people fictional.

Most twins are depth reliefs from one photo, not measured surveys.

Uploads are open. No accounts or moderation yet; one setting can lock writes.

Three models are non-commercial (NLLB-200, Depth Anything V2 Base, Qwen2.5-VL).

Roadmap

Next

- **Managed PostgreSQL and S3,** so uploads persist
- **Accounts, roles and a moderation queue**
- **An always-on GPU worker;** PostgreSQL tested in CI

Then

- **Handwriting OCR fine-tuned** on Gujarati and Devanagari journals
- **All 22 scheduled languages** through IndicTrans2
- **Multi-photo Gaussian splats** as the default twin

Later

- **Partner archives** ingesting real collections
- **IIIF and Dublin Core export**
- **Mobile field capture** with GPS and offline sync

Thank you

આભાર • धन्यवाद

Live	heritagelens-hc26.vercel.app
Code	github.com/kushpatel2601/HeritageLens
Demo	heritagelens-hc26.vercel.app/#demo
API docs	heritagelens-hc26.onrender.com/docs